Compositional Game Theory

An introduction

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Definition 0.1: A **Normal-Form Game** is a tuple $\left(P, S_{p \in P}, u\right)$ where:

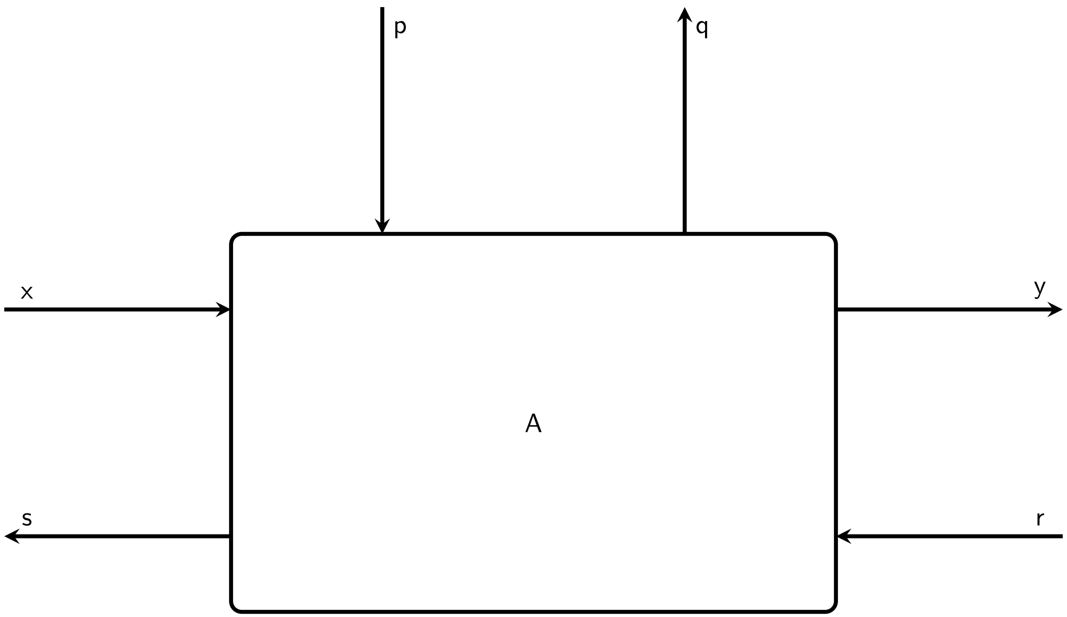
- P is a set of n players
- S_p is the set of player p's strategies
- $u : \prod_{p \in P} S_p \to \mathbb{R}^n$ is the utility function

Definition 0.2: A **Strategy Profile** for a game, is a choice of $s \in S_p$ for all players p

Definition 0.3: A strategy profile s^* is a **Nash Equilibrium** if for all players p, and for all $s_p \in S_p$: $u_i\left(s_p^*, s_{-p}^*\right) \geq u_i\left(s_p, s_{-p}^*\right)$

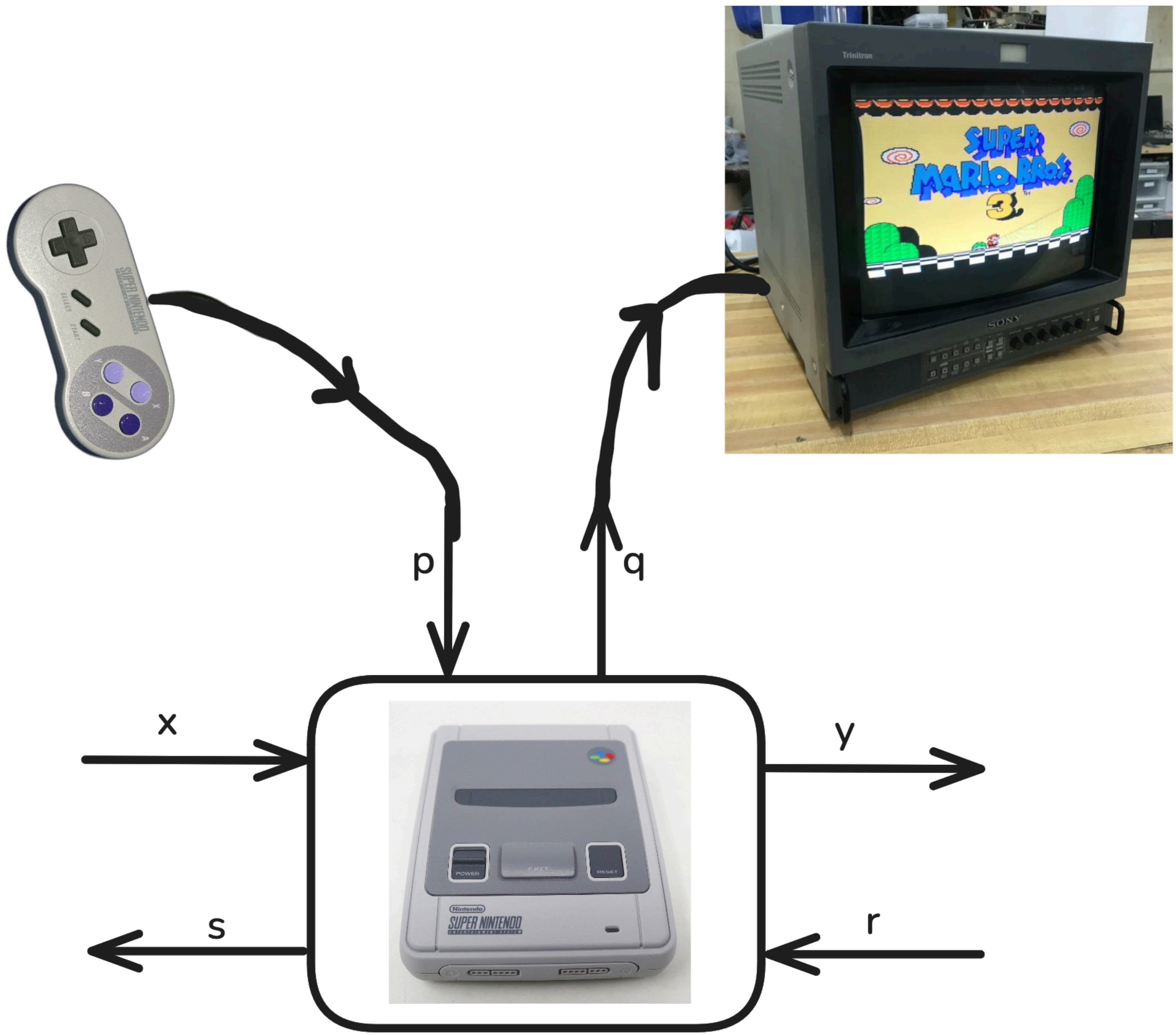
Definition 0.4: An **Open Game** is a pair (A, p), where A is a **Parametrized Lens** with parameters $\binom{P}{Q}$ and p is a (non-parametrized) **Lens** $\binom{K}{K \to \mathrm{Bool}} \to \binom{P}{P \to Q}$

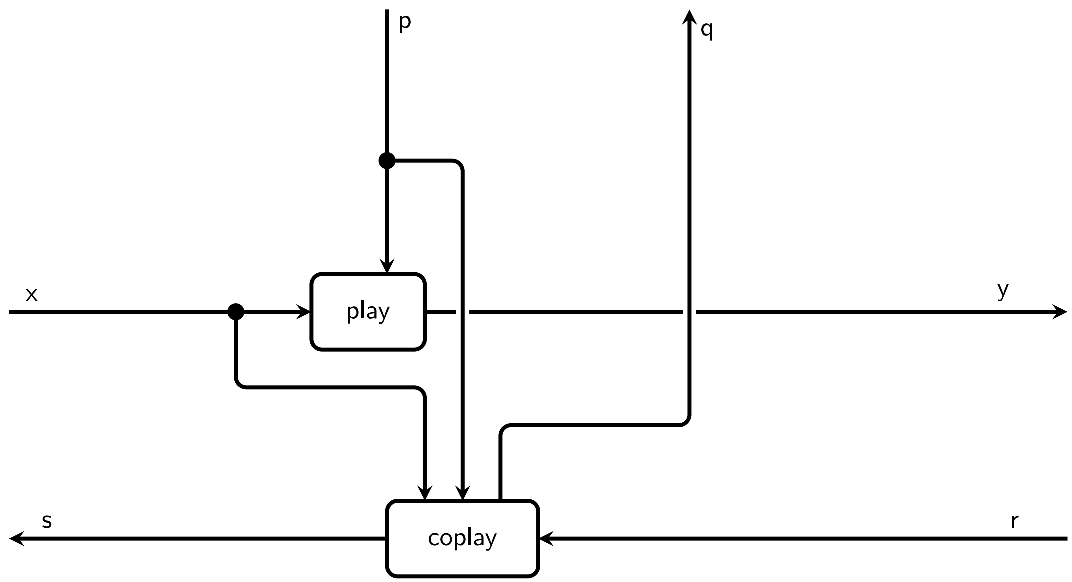
Definition 0.5: A **Parametrized Lens** from a pair of sets $\binom{X}{S}$ to a pair of sets $\binom{Y}{R}$ with **Parameters** $\binom{P}{Q}$ is a pair of functions: $\mathrm{play} : P \times X \to Y$ and $\mathrm{coplay} : P \times X \times R \to S \times Q$



A Parametrized Lens.

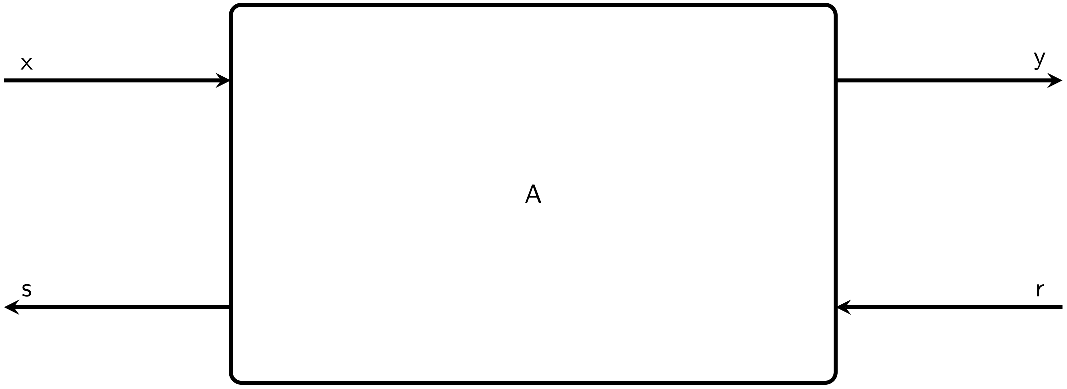
- x: **game states** that can be observed by the player prior to making a move.
- p: **strategies/actions** a player can adopt.
- y: **game states** that can be observed after the player made its move.
- r: **utilities**/**payoffs** the player can receive after making its move.
- s: **back-propagated utilities** a player can send to players that moved before it.
- q: **rewards** representing the player's intrinsic utility.


p
q
x
y
s
r
Trinitron
SONY
SUPER MARIO BROS. 3
POWER
RESET
Nintendo
SUPER NINTENDO

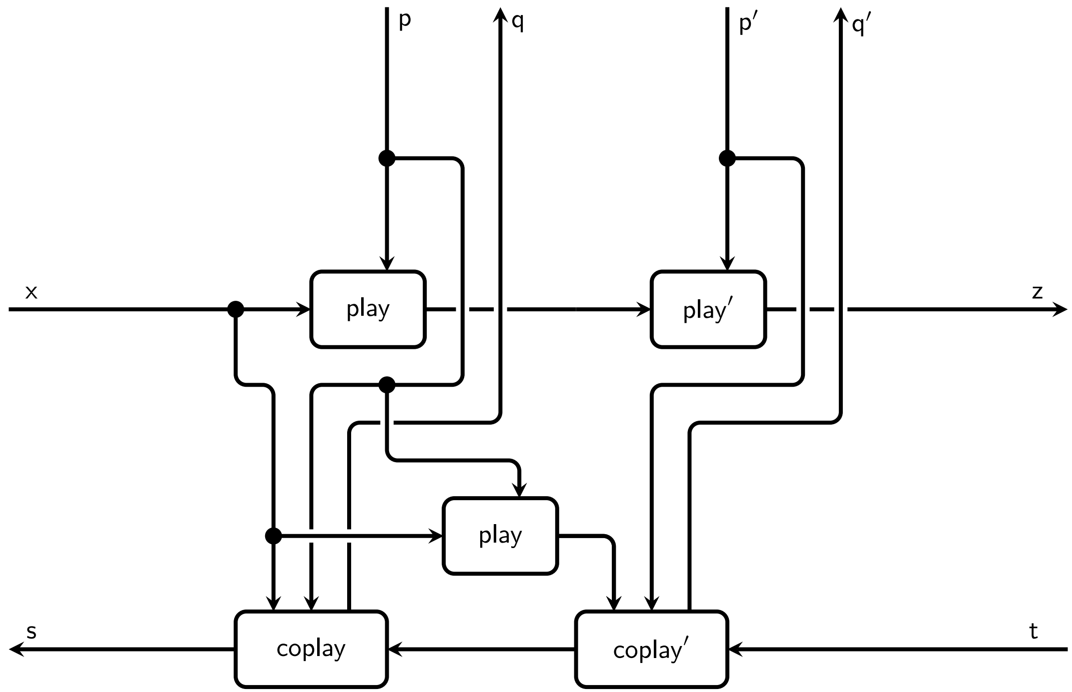


The internals of a Parametrized Lens. Data flows bidirectionally.

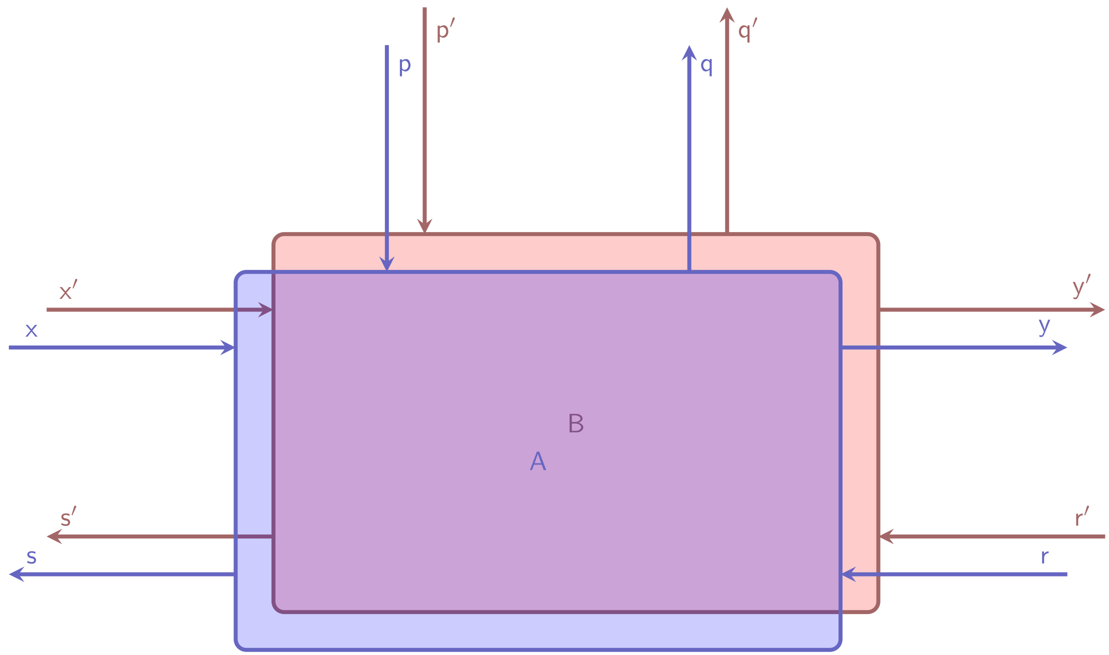
Definition 0.6: A (non-parametrized) **Lens** is just a Parametrized Lens with trivial parameters $\binom{1}{1}$



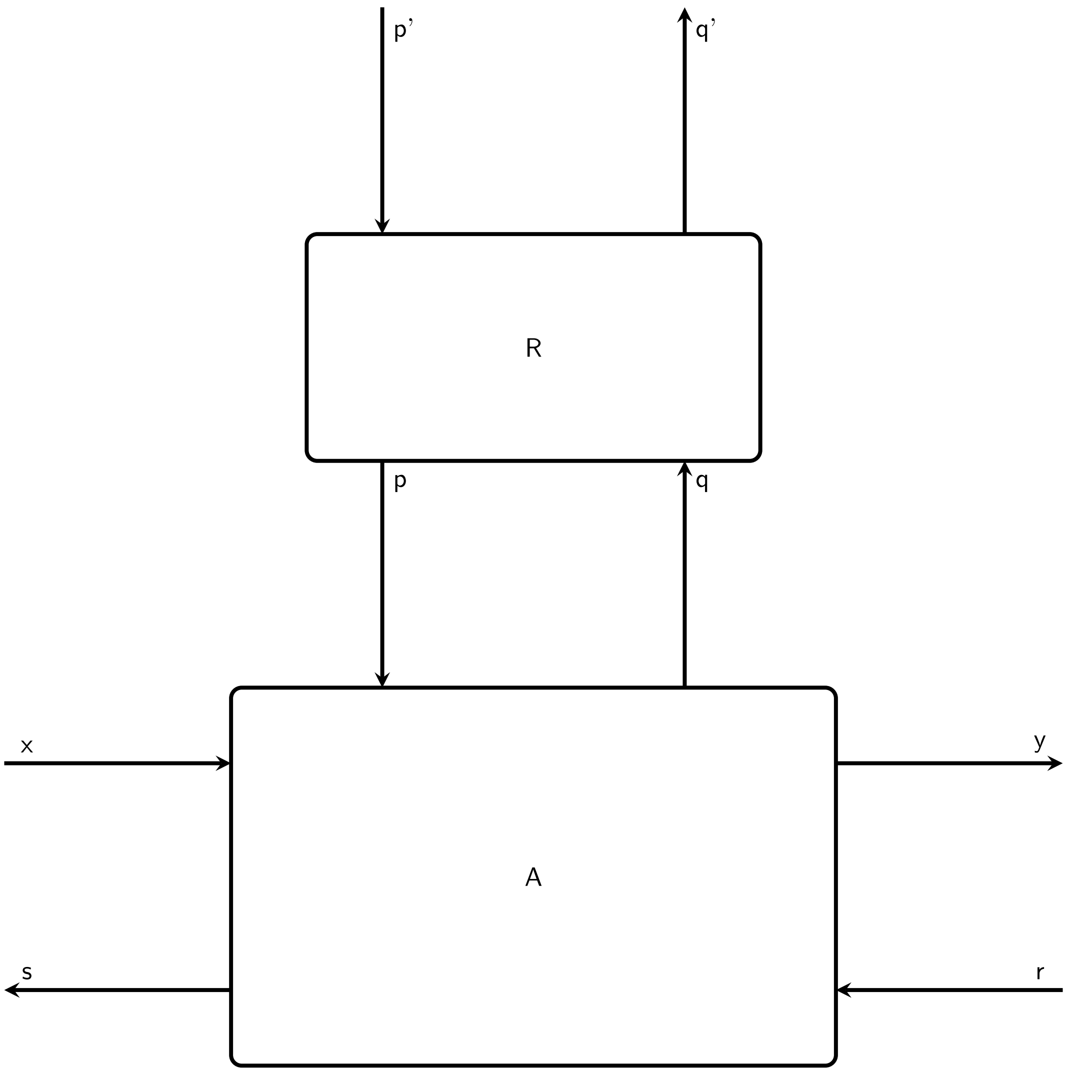
A non-parametrized Lens. Trivial wires are not drawn as they bear no information.



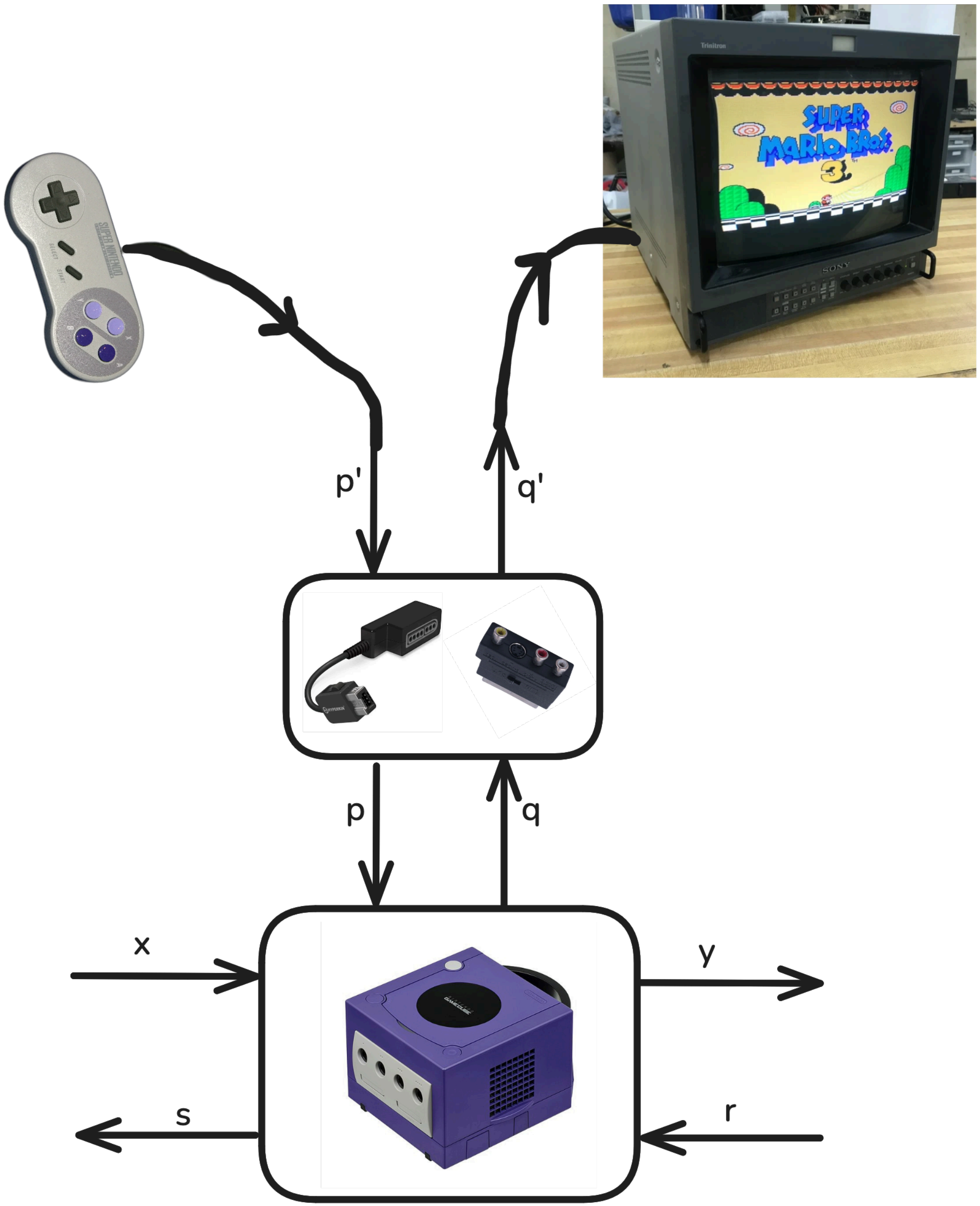
Sequential composition of two lenses with matching types. Both parameters are kept.

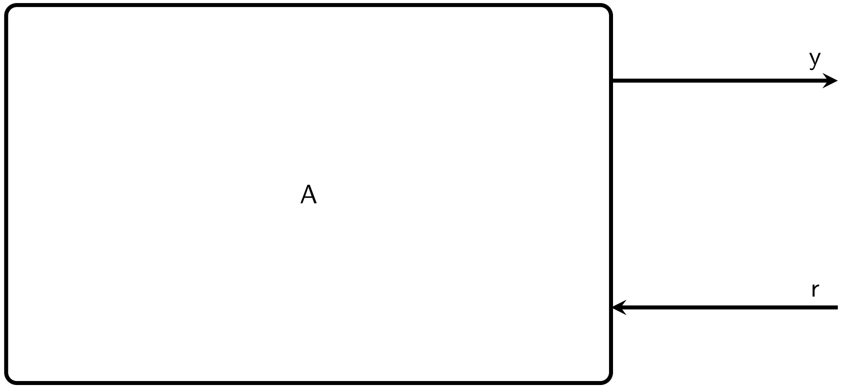


Parallel composition of two lenses. There is no interaction, therefore the boxes are drawn separately.

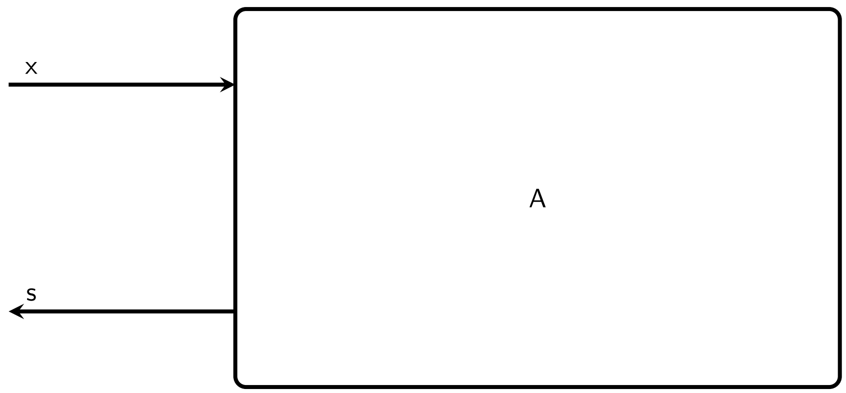


A reparametrization.


p'
q'
p
q
x
y
s
r

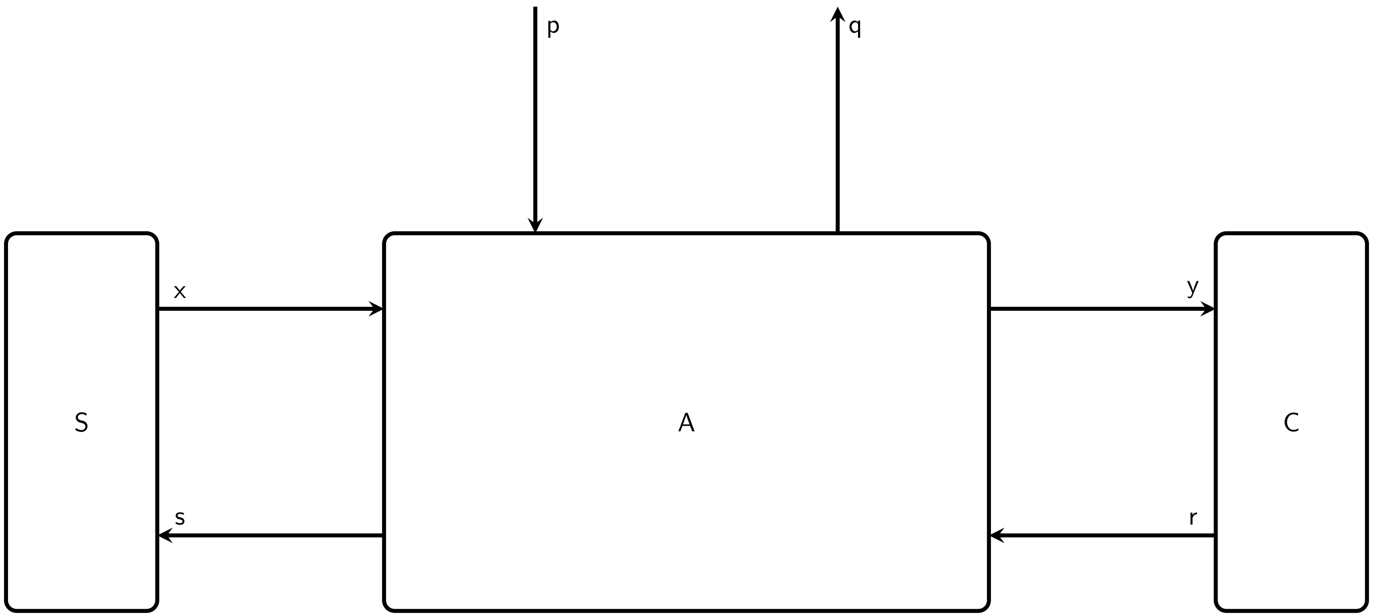


A State, i.e. a non-parametrized Lens with no wires on the left.



A co-State, i.e. a non-parametrized Lens with no wires on the right.

Definition 0.7: An **Environment** for a Parametrized Lens $\begin{pmatrix} X \\ S \end{pmatrix} \to \begin{pmatrix} Y \\ R \end{pmatrix}$ is a pair of a **State** $\begin{pmatrix} X \\ S \end{pmatrix}$ and a **co-State** $\begin{pmatrix} Y \\ R \end{pmatrix}$



A Parametrized Lens enclosed by an Environment.

Port	Games	ML	Controls	DB
p	Strategy	Weights	Control Signal	Schema
q	Utility	Weight Gradient	Cost	Update Log
x	Current State	Input Data	Current State	Source DB
y	Next State	Activations	Next State	New View
r	Payoff	Gradient	Backprop’d Cost	Modified View
s	Backprop’d CoPayoff	Prev. Layer Gradient	Updated Cost-To-Go	Source DB Update